

Matrices and Advanced Calculus

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1 Matrices

1.1 Introduction

1.1.1 Method of solving system of equations with same number of equations and variables

We have already learnt the method of solving this type of system of equations.

$$a_1x_1 + b_1x_2 + c_1x_3 = p$$

$$a_2x_1 + b_2x_2 + c_2x_3 = q$$

$$a_3x_1 + b_3x_2 + c_3x_3 = r$$

To solve this we can assume the following:

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad B = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad \text{and} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Now, we can solve by using $AX = B$ and then taking the inverse of A on both sides. Hence, $X = A^{-1}B$

1.1.2 Less equations, more variables

Consider the equation:

$$x_1 + 2x_2 = 1$$

This is only one equation, but there are two unknowns. So we can let

$$x_2 = \alpha \in \mathbb{R} \implies x_1 = 1 - 2\alpha$$

$$\text{Hence, } X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 - 2\alpha \\ \alpha \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \end{bmatrix} \alpha$$

1.2 Idea of Elimination

1.2.1 Pivots and Multipliers

1. Diagonal positions (11, 22, 33) are called as **pivots** and are **non-zero**.
2. **Multipliers**, $l_{ij} = \frac{Entry to Eliminate}{Pivot}$.
3. We need to multiply the j^{th} equation with l_{ij} and subtract it from the i^{th} equation.

1.2.2 Method of Elimination

1. Consider the following example:

$$x + 2y + 3z = 6$$

$$2x + 5y + 2z = 4$$

$$6x - 3y + z = 2$$

2. Clearly, 1st pivot is 1. $l_{21} = \frac{2}{1} = 2$
3. Now, doing $eq^n(2) - 2 \cdot eq^n(1)$, we get:

$$x + 2y + 3z = 6$$

$$y - 4z = -8$$

$$6x - 3y + z = 2$$

4. Now, $l_{31} = \frac{6}{1} = 6$ and then by $eq^n(3) - 6 \cdot eq^n(1)$, we get:

$$x + 2y + 3z = 6$$

$$y - 4z = -8$$

$$-15y - 17z = -34$$

5. Now, 2nd pivot is 1. So, $l_{32} = \frac{-15}{1}$ and by doing $eq^n(3) + 15 \cdot eq^n(2)$, we get:

$$x + 2y + 3z = 6$$

$$y - 4z = -8$$

$$-77z = -154$$

6. Hence, finally we get $z = 2, y = 0, x = 0$

1.2.3 Elimination by Matrices

1. **Elimination Matrix** (E_{ij}): The matrix which is multiplied to the coefficient matrix to get rid of the coefficients.

$$2. E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -l_{31} & 0 & 1 \end{bmatrix} \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -l_{32} & 1 \end{bmatrix}$$

3. Let's solve the same example by using this method.

$$4. \text{ We have } A = \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 5 & 2 & | & 4 \\ 6 & -3 & 1 & | & 2 \end{bmatrix}. \text{ Also } l_{21} = 2 \implies E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$5. \text{ Now, } A_1 = E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 5 & 2 & | & 4 \\ 6 & -3 & 1 & | & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & -4 & | & -8 \\ 6 & -3 & 1 & | & 2 \end{bmatrix}$$

$$6. \text{ Now, } l_{31} = 6 \implies E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix}$$

$$7. \text{ Hence, } A_2 = E_{31}A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & -4 & | & -8 \\ 6 & -3 & 1 & | & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & -4 & | & -8 \\ 0 & -15 & -17 & | & -34 \end{bmatrix}$$

$$8. \text{ Finally, } l_{32} = -15 \implies E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 15 & 1 \end{bmatrix}$$

$$9. \text{ Hence, } B = E_{32}A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 15 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & -4 & | & -8 \\ 0 & -15 & -17 & | & -34 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & -4 & | & -8 \\ 0 & 0 & -77 & | & -154 \end{bmatrix}$$

10. Finally, we get $z = 2, y = 0, x = 0$

Question:

$$x + 2y - z = 1$$

$$4x + 9y - 3z = 8$$

$$-2x - 3y + 7z = 10$$

$$\text{We have } A = \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 4 & 9 & -3 & | & 8 \\ -2 & -3 & 7 & | & 10 \end{bmatrix} \implies l_{21} = 4$$

$$E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 4 & 9 & -3 & | & 8 \\ -2 & -3 & 7 & | & 10 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 0 & 1 & 1 & | & 4 \\ -2 & -3 & 7 & | & 10 \end{bmatrix}$$

$$E_{31}E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 0 & 1 & 1 & | & 4 \\ -2 & -3 & 7 & | & 10 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 0 & 1 & 1 & | & 4 \\ 0 & 1 & 5 & | & 12 \end{bmatrix}$$

$$E_{32}E_{31}E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 0 & 1 & 1 & | & 4 \\ 0 & 1 & 5 & | & 12 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 & | & 1 \\ 0 & 1 & 1 & | & 4 \\ 0 & 0 & 4 & | & 8 \end{bmatrix}$$

Hence, $z = 2, y = 2, x = -1$

1.3 Inverse Matrices

1. $\exists A^{-1} \implies AA^{-1} = I$
2. If A is a nxn matrix, and A^{-1} exists, then A has **n** pivots

$$3. D = \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ 0 & 0 & d_3 & \dots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix} \implies D^{-1} = \begin{bmatrix} \frac{1}{d_1} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{d_2} & 0 & \dots & 0 \\ 0 & 0 & \frac{1}{d_3} & \dots & 0 \\ \dots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{d_n} \end{bmatrix}$$

4. **Inverse of an Elimination Matrix:**

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.4 Gauss-Jordan Method

1. **Method:** In $[A \mid I]$ Using elimination, make $a_{21}, a_{31}, a_{32}, a_{23}, a_{13}, a_{12}$ positions zero (in that order, for a 3x3 matrix). Then divide each row by its pivot position to get I as the first matrix. The second matrix in the augmented matrix is the inverse.
2. **Hint:** While eliminating using l_{ij} , in the next step, write all rows except i as it is, and then perform $R_i \rightarrow R_i - l_{ij}R_j$ to get the i^{th} row. [Basically multiply $-l_{ij}$ to j^{th} row and add it to the i^{th} row]

$$3. \text{ **Example:** Find inverse of } A = \begin{bmatrix} 2 & -1 & 0 & | & 1 & 0 & 0 \\ -1 & 2 & -1 & | & 0 & 1 & 0 \\ 0 & -1 & 2 & | & 0 & 0 & 1 \end{bmatrix} [l_{21} = -\frac{1}{2}]$$

$$(a) E_{21}A = \begin{bmatrix} 2 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & | & \frac{1}{2} & 1 & 0 \\ 0 & -1 & 2 & | & 0 & 0 & 1 \end{bmatrix} [l_{32} = -\frac{2}{3}]$$

$$(b) E_{32}A = \begin{bmatrix} 2 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & | & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & | & \frac{1}{3} & \frac{2}{3} & 1 \end{bmatrix} [l_{23} = -\frac{3}{4}]$$

$$\begin{aligned}
\text{(c) } E_{23}A &= \left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array} \right] [l_{12} = -\frac{2}{3}] \\
\text{(d) } E_{12}A &= \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & 1 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} & 1 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{array} \right] \\
\text{(e) Hence, } A^{-1} &= \left[\begin{array}{ccc} \frac{3}{4} & 1 & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{array} \right] = \frac{1}{4} \left[\begin{array}{ccc} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{array} \right]
\end{aligned}$$

1.5 LU Decomposition

1. We need to factorize $A = LU$
2. We know that $E_{32}E_{31}E_{21}A = U \Rightarrow A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U$
 $\Rightarrow L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$
3. $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_{31} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & l_{32} & 1 \end{bmatrix} \Rightarrow L = \boxed{\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}}$
4. To solve $AX = B$, let $A = LU \Rightarrow (LU)x = B \Rightarrow L(UX) = B$
5. Let $UX = Y \Rightarrow LY = B \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$
6. $y_1 = b_1 \Rightarrow l_{21}y_1 + y_2 = b_2 \Rightarrow l_{31}y_1 + l_{32}y_2 + y_3 = b_3$
7. After Finding Y, we can easily find X.

1.5.1 Question:

Solve using LU Decomposition:

$$x + 2y - z = 1$$

$$4x + 9y - 3z = 8$$

$$-2x - 3y + 7z = 10$$

$$\text{We have } A = \begin{bmatrix} 1 & 2 & -1 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \Rightarrow l_{21} = 4$$

$$E_{21}A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{bmatrix} \Rightarrow l_{31} = -2$$

$$E_{31}E_{21}A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{bmatrix} \implies l_{32} = 1$$

$$U = E_{32}E_{31}E_{21}A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 4 \end{bmatrix} \implies L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix}$$

$$LY = B \implies \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 10 \end{bmatrix} \implies y_1 = 1, y_2 = 4, y_3 = 8$$

$$UX = Y \implies \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix} \implies \boxed{x = -1, y = 2, z = 2}$$

1.6 Vector Space

1. A space $\mathbb{R}^n$ consisting a set of vectors is a vector space such that

$$\mathbb{R}^n = \left[\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \middle| v_1, v_2, \dots, v_n \in \mathbb{R} \right]$$

2. The vectors in these spaces are not necessarily column vectors.
3. In the definition of vector space, vector addition ($u+v$) and scalar multiplication (cu) must follow the following **eight rules**:

- (a) $u+v=v+u$
- (b) $u+(v+w) = u+(v+w)$
- (c) $\exists$ unique zero vector s.t. $u+0=u \forall u$
- (d) $\exists$ unique additive inverse $-u$ s.t. $u+(-u)=0 \forall u$
- (e) $\exists$ multiplicative identity s.t. $1 \cdot u = u \forall u$
- (f) $(c_1 \cdot c_2)u = c_1 \cdot (c_2u)$
- (g) $c(u+v) = c \cdot u + c \cdot v$
- (h) $(c_1 + c_2) \cdot u = c_1u + c_2u$

4. **Question:** Let V be a set of polynomials with real coefficients of degree ' n '. Check if it is a vector space w.r.t addition and scalar multiplication:

$$V = \mathbb{P}_n(x) = \{P_0 + P_1x + P_2x^2 + \dots + P_nx^n | P_i \in \mathbb{R}\}$$

Let

$$a_n(x) = \{1 + x + x^2 + \dots + x^{n-1} + x^n\}$$

$$b_n(x) = \{1 + x + x^2 + \dots + x^{n-1} - x^n\}$$

For V to be a vector space, $a_n(x) + b_n(x)$ must be in V . But,

$$a_n(x) + b_n(x) = \{2 + 2x + 2x^2 + \dots + 2x^{n-1} + 0x^n\} \notin V$$

as $a_n(x) + b_n(x)$ is now of degree $n-1$, instead of n .

5. V is a set of polynomials of **degree less than or equal to 'n'** forms a **vector space**.

6. **Question:** $V = \mathbb{P}_n$ is a set of polynomial of degree 'n' where $u+v=u \cdot v$. Check if V forms a vector space or not:

Let u and v be polynomials of degree 'n' such that $u, v \in V$. For V to be a vector space, $u + v \in V \implies u \cdot v \in V$. But $u \cdot v$ is a polynomial of degree '2n'. Hence $u \cdot v \notin V$, which means V is not a vector space.

7. **Question:** Check if $V = \{(x, y) | x, y \in \mathbb{R}\}$ is a vector space with respect to addition and scalar multiplication where

$$u + v = (x_1, y_1) + (x_2, y_2) = (x_1x_2, y_1y_2)$$

$$\alpha u = \alpha(x_1, y_1) = (\alpha x_1, \alpha y_1)$$

$$(a) \quad u + v = (x_1, y_1) + (x_2, y_2) = (x_1x_2, y_1y_2) = (x_2x_1, y_2y_1) = (x_2, y_2) + (x_1, y_1) = v + u$$

$$(b) \quad u + (v + w) = (x_1, y_1) + ((x_2, y_2) + (x_3, y_3)) = (x_1, y_1) + (x_2x_3, y_2y_3) = (x_1(x_2x_3), y_1(y_2y_3)) = ((x_1x_2)x_3, (y_1y_2)y_3) = ((x_1x_2, y_1y_2) + (x_3, y_3)) = ((x_1, y_1) + (x_2, y_2)) + (x_3, y_3) = (u + v) + w$$

$$(c) \quad \exists (1, 1) \in \mathbb{R} \implies (x_1, y_1) + (1, 1) = (x_1, y_1) = u \forall u$$

$$(d) \quad \exists (-1, -1) \in \mathbb{R} \implies (x_1, y_1) + (-x_1, -y_1) = (-x_1^2, -x_2^2) \neq (1, 1)$$

Hence, as property #4 fails, V is not a vector space.

1.6.1 Sub Space:

1. A subspace of a vector space is a set of vectors (including 0) that satisfies the following two requirements, if u and v are vectors in the subspace and α is any scalar, then:

$$(a) \quad u+v \text{ is in the subspace } (u + v \in W)$$

$$(b) \quad \alpha u \text{ is in the subspace } (\alpha u \in W)$$

2. If W is the subspace and V is the vector space, then $W \subset V$

3. **Question:** If $V = \mathbb{R}^2$ and $W = \{(x, y) | x \geq 0, y \geq 0\}$. Check if W is a subspace of V or not.

$$(a) \quad u, v \in W \implies u + v = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in W$$

$$(b) \quad \alpha u = \alpha(x_1, y_1) = (\alpha x_1, \alpha y_1)$$

$$\text{But if } \alpha < 0 \implies \alpha x_1, \alpha y_1 < 0 \implies (\alpha x_1, \alpha y_1) \notin W \implies \alpha u \notin W$$

Hence, W is not a subspace of V .

4. If $V = \mathbb{R}^3$, check if $P = \{(x, y, z) | ax + by + cz = 0\}$ is a subspace of V .

$$\text{Let } u, v \in P \implies u = (x_1, y_1, z_1), v = (x_2, y_2, z_2)$$

$$\therefore u + v = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$\text{and } \alpha u = \alpha \cdot (x_1, y_1, z_1) = (\alpha x_1, \alpha y_1, \alpha z_1)$$

$$\text{Consider } a(x_1 + x_2) + b(y_1 + y_2) + c(z_1 + z_2)$$

$$\implies (ax_1 + by_1 + cz_1) + (ax_2 + by_2 + cz_2) = 0 + 0 = 0 \implies u + v \in P$$

$$\text{Consider } a \cdot (\alpha x_1) + b \cdot (\alpha y_1) + c \cdot (\alpha z_1)$$

$$\implies \alpha \cdot (ax_1 + by_1 + cz_1) = 0 \implies \alpha u \in P$$

$$\text{As } u + v \in P \text{ and } \alpha u \in P \implies P \subset V$$

1.6.2 Column Space:

1. The column space consists of all the linear combinations of all columns.
2. It is denoted by " $C(A)$ ".
- 3.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \implies C(A) = \left\{ \alpha \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} + \beta \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} \mid \alpha, \beta \in \mathbb{R} \right\}$$

4. The system $AX = B$ is solvable **if and only if** $B \in C(A)$.
5. If A is a $m \times n$ matrix then $C(A)$ is a subspace of $\mathbb{R}^m$.

1.6.3 Null Space:

1. The null space of A consists all the solutions to the system $AX = 0$.
2. It is denoted by " $N(A)$ ".
3. $N(A) = \{X | AX = 0\}$.
4. If A is a $m \times n$ matrix then $C(A)$ is a subspace of $\mathbb{R}^n$.

1.6.4 Finding null space of a singular matrix:

1. For finding the null space, we need to eliminate the 21, 31 positions as usual using the 1st pivot (11).
2. Now, if the 22 and 23 positions are both zero, meaning that 22 position can not have any pivot, then we call that column "free column".
3. If 22 doesn't yield any pivots, we go to position 32 (right of 22) and if there exists a non-zero value there, we call that a pivot.
4. The columns in which pivots exists are called "pivoted columns".
5. To find the null space of a given singular matrix, we need to determine the free and pivoted columns using matrix elimination.
6. The variables in the X matrix $(x_1, x_2, \dots, x_n)$ corresponding to the free columns are called "free variables" and those corresponding to the pivoted columns are called "pivoted variables".
7. We let one of the free variables equal to 1 and all the other free variables equal to zero. Now, substituting the free variables in the equation formed by $UX = 0$ yields the pivoted variables. And the vector thus formed $(x_1, x_2, \dots, x_n)$ is labelled as S_1 .
8. Repeat this process for all the free variables letting one 1 and the rest 0.
9. The set of all S makes $N(A)$.

10. **Example:** Find null space of $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & 2 & 8 & 10 \\ 3 & 3 & 10 & 13 \end{bmatrix}$

$$E_{21}A = \begin{bmatrix} \boxed{1} & 1 & 2 & 3 \\ 0 & 0 & 4 & 4 \\ 3 & 3 & 10 & 13 \end{bmatrix} \Rightarrow E_{31}A_1 = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 0 & \boxed{4} & 4 \\ 0 & 0 & 4 & 4 \end{bmatrix} \Rightarrow E_{33}A_2 = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

By $UX=0$, we get $x_1 + x_2 + 2x_3 + 3x_4 = 0$ and $4x_3 + 4x_4 = 0$
 $\Rightarrow x_3 = -x_4$ and $x_1 + x_2 + x_4 = 0$

Here, x_2 and x_4 are free variables.

Case 1: $x_2 = 1, x_4 = 0 \Rightarrow x_1 = -1, x_3 = 0 \Rightarrow S_1 = (-1, 1, 0, 0)$

Case 2: $x_2 = 0, x_4 = 1 \Rightarrow x_1 = -1, x_3 = -1 \Rightarrow S_1 = (-1, 0, -1, 1)$

$$\therefore N(A) = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right\}$$

1.7 Row Reduced Echlon Form (RREF):

1. To get the **row reduced echlon form** of a matrix A, denoted by $rref(A)$, we make all the entries above the pivots of U zero.
2. To get the values of special solutions S for the null space of A, we let each of the free variables as 1 and make the rest 0, as we did above.
3. The pivoted variable spots in the special solution, will be the opposite sign of the column entries of the corresponding free variable columns.
4. **Rank:** The number of pivots in RREF of a matrix is its rank.

5. **Question:** $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}$ Find the rank(A), C(A), N(A).

$$E_{21}A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 1 & 3 & 2 & 6 \end{bmatrix} \Rightarrow E_{31}A_1 = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix} \Rightarrow E_{32}A_2 = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow rref(A) = E_{12}U = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence, rank(A) = 2

Now, as x_2, x_4 are free variables, for first special solution, x_1, x_3 are going to be opposite sign of the entries in the x_2 column. Similarly, for second special solution, they will be opposite sign of entries in the x_4 column.

$$N(A) = S_1, S_2 = \left\{ \begin{pmatrix} -2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Column Space is the columns of the "A" which have pivots in the RREF.

$$C(A) = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \right\}$$

1.7.1 Complete solution of $AX = B$:

1. For a system of equations $AX = B$, the complete solutions, $X = X_N + X_p$ where X_N is the linear combination of null space $AX = 0$ and X_p comes from the right-hand side vector
2. Firstly, we convert A into the RREF such that $[A \mid B] \rightarrow [R \mid d]$
3. For X_p , the free variable corresponding row will be zero, and the rest of the entries will come directly from the "d" vector with the same sign.

4. **Question:** $\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 1 & 3 & 1 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix}$. For system $AX = B$, find the complete set of solutions.

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 1 & 3 & 1 & 6 & 7 \end{array} \right], l_{31} = 1 \Rightarrow \left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 4 & 6 \end{array} \right], l_{33} = 1 \Rightarrow \left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$S_1 = \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, S_2 = \begin{pmatrix} -2 \\ 0 \\ -4 \\ 1 \end{pmatrix} \Rightarrow X_N = \alpha \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ 0 \\ -4 \\ 1 \end{pmatrix}, X_p = \begin{pmatrix} 1 \\ 0 \\ 6 \\ 0 \end{pmatrix}$$

$$\text{Hence, } X = X_N + X_p = \alpha \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ 0 \\ -4 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 6 \\ 0 \end{pmatrix}$$

1.7.2 Observation for certain matrices:

Consider m is the number of rows in a matrix, n is the number of columns in a matrix, and r is the rank of the matrix:

1. **Tall Matrix:** Full column rank, $m > n$, $n = r$, zero / unique solution.
2. **Wide Matrix:** Full row rank, $m < n$, $m = r$, infinite solutions.
3. **Square Matrix:** Full rank matrix, $m = r$, $n = r$, unique solution.
4. **Not Full Matrix:** $r < m$, $r < n$, zero / infinite solutions.

1.7.3 Question:

Find for what values of λ , the equation $x + y + z = 1$; $x + 2y + 4z = \lambda$; $x + 4y + 10z = \lambda^2$ have a solution and solve them completely in each λ .

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & \lambda \\ 1 & 4 & 10 & \lambda^2 \end{array} \right] \xrightarrow{l_{21} \Rightarrow 1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \lambda - 1 \\ 1 & 4 & 10 & \lambda^2 \end{array} \right] \xrightarrow{l_{31} \Rightarrow 1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \lambda - 1 \\ 0 & 3 & 9 & \lambda^2 - 1 \end{array} \right]$$

$$\xrightarrow{l_{32} \Rightarrow 3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \lambda - 1 \\ 0 & 0 & 0 & \lambda^2 - 3\lambda + 2 \end{array} \right] \xrightarrow{l_{12} \Rightarrow 1} \text{rref}(A) = \left[\begin{array}{ccc|c} 1 & 0 & -2 & 2 - \lambda \\ 0 & 1 & 3 & \lambda - 1 \\ 0 & 0 & 0 & \lambda^2 - 3\lambda + 2 \end{array} \right]$$

For solution, $\lambda^2 - 3\lambda + 2 = 0 \Rightarrow (\lambda - 2)(\lambda - 1) = 0 \Rightarrow \lambda = 1, 2$

$$X_N = \alpha \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, X_p = \begin{pmatrix} 2 - \lambda \\ \lambda - 1 \\ 0 \end{pmatrix} \Rightarrow X = \alpha \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 - \lambda \\ \lambda - 1 \\ 0 \end{pmatrix}$$

1.8 Some Important Definitions:

1.8.1 Linear Independency:

1. Vectors $\{X_1, X_2, X_3, \dots, X_n\}$ are linearly independent iff
$$\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \dots + \alpha_n X_n = 0 \implies \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$$
2. The columns of A are linearly independent when the solution to the system $AX = 0$ is $X = 0$.

1.8.2 Linear Dependency:

Vectors $\{X_1, X_2, X_3, \dots, X_n\}$ are linearly dependent iff

$$\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \dots + \alpha_n X_n = 0 \implies \exists \text{ at least one } \alpha_i \neq 0$$

1.8.3 Span:

1. A set of vectors span a space if the linear combination fill the entire space.
2. **Column Space:** The column space of a matrix A is the subspace of $V = \mathbb{R}^m$ spanned by the columns, denoted by $C(A)$
3. **Row Space:** The row space of a matrix A is the subspace of $V = \mathbb{R}^n$ spanned by the rows, denoted by $C(A^T)$
4. **Left Null Space:** The set of solutions of $A^T X = 0$, denoted by $N(A^T)$

1.8.4 Basis:

A basis for a vector space is the sequence of vectors with two properties:

1. Vectors are linearly independent.
2. They span the space.

1.8.5 Dimensions:

The dimension of a space is the number of vector in its basis.

1.8.6 Some Important Points:

1. $C(A) \perp N(A^T)$ and $C(A^T) \perp N(A)$ as their dot products are zero.
2. For $A_{m \times n}$, $\dim(C(A)) = \dim(C(A^T)) = r$,
 $\dim(N(A)) = n - r$, $\dim(N(A^T)) = m - r$
3. If in $\text{rref}(A)$, an entire row is zero, then the left null space will be the corresponding rows in the total elimination matrix.

4. **Example:** Find the four subspaces of $A = \begin{bmatrix} 1 & 3 & 0 & 5 \\ 2 & 6 & 1 & 16 \\ 5 & 15 & 0 & 25 \end{bmatrix}$ ($l_{21} = 2$)

$$E_{21}A = \begin{bmatrix} 1 & 3 & 0 & 5 \\ 0 & 0 & 1 & 6 \\ 5 & 15 & 0 & 25 \end{bmatrix} (l_{31} = 5) \implies rref(A) = \begin{bmatrix} \boxed{1} & 3 & 0 & 5 \\ 0 & 0 & \boxed{1} & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}, E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$$

$$\therefore C(A) = \left\{ \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, C(A^T) = \left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 6 \end{pmatrix} \right\},$$

$$N(A) = \left\{ \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -5 \\ 0 \\ -6 \\ 1 \end{pmatrix} \right\}, N(A^T) = \left\{ \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} \right\}$$

5. **Question:** Find the four subspaces of $A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 5 \end{bmatrix}$ ($l_{31} = 1$)

$$E_{31}A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} (l_{32} = 1) \implies E_{32}A_1 = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} (l_{12} = 1)$$

$$rref(A) = E_{12}A_2 = \begin{bmatrix} \boxed{1} & 0 & -1 & -1 \\ 0 & \boxed{1} & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies E = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\therefore C(A) = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\}, C(A^T) = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix} \right\},$$

$$N(A) = \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\}, N(A^T) = \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

1.9 Orthogonality

For two vectors u and v to be orthogonal, $u \cdot v = 0$. In matrix form, $u^T v = 0$. As u , and v are perpendicular, u , v , and $u+v$ form a right triangle. By pythagoras theorem,

$$||u||^2 + ||v||^2 = ||u + v||^2$$

$$u^T u + v^T v = u^T u + v^T v + u^T v + v^T u \implies u^T v + v^T u = 0$$

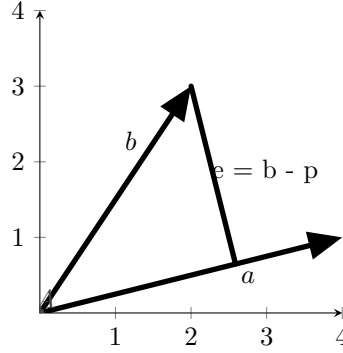
But, we know that $u^T v = v^T u \implies \boxed{u^T v = 0 \implies u \cdot v = 0}$

Hence, proved that u and v are orthogonal to each other.

1.9.1 Projection:

- error, $e = b - p \implies e = b - \hat{x}a$
 Now, $a \cdot e = 0 \implies a^T e = 0$
 $\therefore a^T(b - \hat{x}a) = 0 \implies a^T b - a^T a \hat{x} = 0$

$$\hat{x} = \frac{a^T b}{a^T a} \implies \boxed{p = \left(\frac{a^T b}{a^T a} \right) a = Pa}$$



- In case of a plane, vector a is resolved into two components a_1 and a_2 such

$$\text{that } A = \begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix} \implies P = A\hat{X} = a_1\hat{x}_1 + a_2\hat{x}_2 = \begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$$

$$\text{Now, } e = b - p = b - A\hat{X}$$

$$\therefore a_1(b - A\hat{X}) = 0 \text{ and } a_2(b - A\hat{X}) = 0 \implies \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} (b - A\hat{X}) = 0$$

$$\therefore A^T(b - A\hat{X}) = 0 \implies A^T b - A^T A \hat{X} = 0 \implies (A^T A) \hat{X} = A^T b$$

$$\hat{X} = (A^T A)^{-1} A^T b \implies \boxed{p = A(A^T A)^{-1} A^T b = Pb}$$

- Properties of Projection Matrix (P):** $P^T = P$ and $P^2 = P$

- Solving $AX=b$ with no solutions:**

If $b \in C(A) \implies$ solution exists, but if $b \notin C(A) \implies$ no solution exist

Replace b by p to get an approximate solution $\hat{X}$

$$\implies AX = b \rightarrow A\hat{X} = p \text{ where } p \in C(A)$$

1.9.2 Problems based on projection:

1. **Question:** Find the closest line to the points (0,6), (1,0), (2,0).

Let the equation be $b = c + Dt$

$$(0, 6) \implies t = 0 \implies c + 0D = 6$$

$$(1, 0) \implies t = 1 \implies c + 1D = 0$$

$$(2, 0) \implies t = 2 \implies c + 2D = 0$$

In matrix form,
$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c \\ D \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 6 \\ 1 & 1 & 0 \\ 1 & 2 & 0 \end{array} \right] \left(\begin{array}{l} l_{21} = 1 \\ l_{31} = 1 \end{array} \right) \implies \left[\begin{array}{cc|c} 1 & 0 & 6 \\ 0 & 1 & -6 \\ 0 & 2 & -6 \end{array} \right] (l_{32} = 2) \implies \left[\begin{array}{cc|c} 1 & 0 & 6 \\ 0 & 1 & -6 \\ 0 & 0 & 6 \end{array} \right]$$

Clearly, from the last row, there exists no solution for the given equation.

Now, we know $(A^T A)\hat{X} = A^T b$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \hat{X} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \hat{X} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & 3 & 6 \\ 3 & 5 & 0 \end{array} \right] (l_{21} = 1) \implies \left[\begin{array}{cc|c} 3 & 3 & 6 \\ 0 & 2 & -6 \end{array} \right] \implies \left(\begin{array}{l} 2\hat{x}_2 = -6 \implies \hat{x}_2 = -3 \\ 3\hat{x}_1 - 9 = 6 \implies \hat{x}_1 = 5 \end{array} \right)$$

Hence, the required line is $\boxed{y = 5 - 3x}$

Also,

$$p = \hat{x}_1 a_1 + \hat{x}_2 a_2 = 5 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 3 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \implies \boxed{p = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}}$$

and error,

$$e = b - p = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} \implies \boxed{e = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}}$$

2. **Question:** Find the closest line to the points (1,1), (2,2), (3,2).

Let the equation be $b = c + Dt$

$$(1,1) \implies t = 1 \implies c + 1D = 1$$

$$(2,2) \implies t = 2 \implies c + 2D = 2$$

$$(3,2) \implies t = 3 \implies c + 3D = 2$$

Clearly, no solution exists.

In matrix form,
$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Now, we know $(A^T A) \hat{X} = A^T b$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \hat{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \implies \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \hat{X} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & | & 5 \\ 6 & 14 & | & 11 \end{bmatrix} (l_{21} = 2) \implies \begin{bmatrix} 3 & 6 & | & 5 \\ 0 & 2 & | & 1 \end{bmatrix} \implies \begin{pmatrix} 2\hat{x}_2 = 1 \implies \hat{x}_2 = 1/2 \\ 3\hat{x}_1 + 3 = 5 \implies \hat{x}_1 = 2/3 \end{pmatrix}$$

Hence, the required line is $\boxed{y = 0.67 + 0.5x}$

1.10 Orthonormal Vectors:

1. $q_1, q_2, q_3, \dots, q_n$ are said to be "orthonormal vectors", if:

$$q_i \cdot q_j = q_i^T q_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

2. For normalizing an orthogonal vector "a" into the orthonormal form "q",

we divide the vector by its norm. $\boxed{q_i = \frac{a_i}{||a_i||}}$

3. **Orthonormal Matrix:** $Q = \begin{pmatrix} | & | & \dots & | \\ q_1 & q_2 & \dots & q_n \\ | & | & \dots & | \end{pmatrix}$

$$\therefore Q^T Q = \begin{pmatrix} - & q_1 & - \\ - & q_2 & - \\ \dots & \dots & \dots \\ - & q_n & - \end{pmatrix} \begin{pmatrix} | & | & \dots & | \\ q_1 & q_2 & \dots & q_n \\ | & | & \dots & | \end{pmatrix} = \begin{pmatrix} q_1^T q_1 & q_1^T q_2 & \dots & q_1^T q_n \\ q_2^T q_1 & q_2^T q_2 & \dots & q_2^T q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \dots & q_n^T q_n \end{pmatrix}$$

$$\therefore Q^T Q = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \implies \boxed{Q^T Q = I}$$

Post multiplying with $Q^{-1} \implies \boxed{Q^T = Q^{-1}}$

4. If $AX = b$ is not solvable, replace A with Q .

$$\therefore AX = b \implies QX = b \implies Q^T QX = Q^T b \implies \boxed{\hat{X} = Q^T b}$$

$$\text{Also, } p = A\hat{X} \implies p = Q\hat{X} = q_1\hat{x}_1 + q_2\hat{x}_2 + \dots + q_n\hat{x}_n$$

$$\text{Hence, } P = A(A^T A)^{-1} A^T = Q(Q^T Q)^{-1} Q^T \implies \boxed{P = QQ^T}$$

5. **Question:** If $Q = \frac{1}{3} \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}$, $b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, find $\hat{X}, p, P, e$

$$\hat{X} = Q^T b = \frac{1}{3} \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$p = q_1\hat{x}_1 + q_2\hat{x}_2 + q_3\hat{x}_3 = \frac{2}{9} \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \frac{2}{9} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} - \frac{1}{9} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$P = QQ^T = I$$

$$e = b - p = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

1.11 Gram-Schmidt Process

1. If a, b, c are independent vectors and A, B, C are orthogonal vectors then:

$$\text{Let } \boxed{A = a}$$

$$A \perp B \text{ and } B = e = b - p$$

$$\implies \boxed{B = b - \hat{x}a = b - \frac{A^T b}{A^T A} A}$$

Now,

$$A \perp C \text{ and } B \perp C \implies \boxed{C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B}$$

2. **Generalization:** If $v_1, v_2, \dots, v_k, \dots, v_n$ are n independent vectors and $u_1, u_2, \dots, u_k, \dots, u_n$ are the corresponding orthogonal vectors, then:

$$\boxed{u_k = v_k - \frac{u_1^T v_k}{u_1^T u_1} u_1 - \frac{u_2^T v_k}{u_2^T u_2} u_2 - \dots - \frac{u_{k-1}^T v_k}{u_{k-1}^T u_{k-1}} u_{k-1}}$$

3. **Question:** Let independent vectors $a = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $b = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix}$, $c = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$.

Find all orthonormal vectors using gram-schmidt process.

$$A = a = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$B = b - \frac{A^T b}{A^T A} A = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} - \frac{2}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \Rightarrow B = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} - \frac{6}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - \frac{-6}{6} \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} \Rightarrow C = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Hence,

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, q_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, q_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

1.11.1 QR Factorization:

$$1. A = QR \Rightarrow Q^{-1}A = R \Rightarrow R = Q^T A$$

2.

$$R = Q^T A = \begin{bmatrix} q_1^T \\ q_2^T \\ q_3^T \end{bmatrix} \begin{bmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} q_1^T a_1 & q_1^T a_2 & q_1^T a_3 \\ q_2^T a_1 & q_2^T a_2 & q_2^T a_3 \\ q_3^T a_1 & q_3^T a_2 & q_3^T a_3 \end{bmatrix}$$

Hence,

$$R = \begin{bmatrix} q_1^T a_1 & q_1^T a_2 & q_1^T a_3 \\ 0 & q_2^T a_2 & q_2^T a_3 \\ 0 & 0 & q_3^T a_3 \end{bmatrix}$$

3. **Question:** If $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{bmatrix}$. Factorize A into QR.

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \Rightarrow R = Q^T A = \begin{bmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{6} & -\sqrt{6} \\ 0 & 0 & \sqrt{3} \end{bmatrix}$$

1.12 Eigenvalues and Eigenvectors:

1. **Eigenvector:** A vector whose transformed vector has the same direction as of the original vector.
2. **Eigenvalue:** The multiple with which the original vector is multiplied to get the transformed vector.
3. **Example:**

$$u = 5x + 4y$$

$$v = x + 2y$$

For

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \implies \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

As these vectors are not in same direction, it is not an eigenvector.

For

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \implies \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \begin{pmatrix} x \\ y \end{pmatrix}$$

As these vectors are in the same direction, it is an eigenvector and eigenvalue is 1.

For

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \implies \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 24 \\ 6 \end{pmatrix} = 6 \begin{pmatrix} x \\ y \end{pmatrix}$$

As these vectors are in the same direction, it is an eigenvector and eigenvalue is 6.

4. In general, $AX = \lambda X$ if λ is the eigenvalue and X is the eigenvector.

$$\therefore AX - \lambda X = 0 \implies (A - \lambda I)X = 0$$

Taking determinant, $|A - \lambda I||X| = 0$

But as $X \neq 0 \implies \boxed{|A - \lambda I| = 0}$

5. **Characteristic Equation:** " $|A - \lambda I| = 0$ " called as the "characteristic equation" of the matrix A .

It is a polynomial equation of degree n and have n roots as:

$$|A - \lambda I| = 0 \implies \lambda^n + p_1\lambda^{n-1} + \dots + p_{n-1}\lambda + p_n = 0$$

The roots of the characteristic equation are the "eigenvalues".

6. **Properties of CEq for 2x2 matrices:**

(a) $\lambda^2 - \lambda(\text{tr}(A)) + \det(A) = 0$

(b) Sum of eigenvalues: $\lambda_1 + \lambda_2 = \text{tr}(A)$

(c) Product of eigenvalues: $\lambda_1\lambda_2 = \det(A)$

7. **Question:** Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$

$$|A - \lambda I| = 0 \implies \begin{vmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

$$\implies (5 - \lambda)(2 - \lambda) - 4 = 0 \implies \lambda^2 - 7\lambda + 6 = 0 \implies \boxed{\lambda = 1, 6}$$

Now,

$$(A - \lambda_1 I)X_1 = 0 \implies \begin{bmatrix} 5 - 1 & 4 \\ 1 & 2 - 1 \end{bmatrix} X_1 = 0 \implies \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} X_1 = 0$$

$$X_1 = \begin{bmatrix} 4 & 4 & | & 0 \\ 1 & 1 & | & 0 \end{bmatrix} = \begin{bmatrix} 4 & 4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \implies X_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Also,

$$(A - \lambda_2 I)X_2 = 0 \implies \begin{bmatrix} 5 - 6 & 4 \\ 1 & 2 - 6 \end{bmatrix} X_2 = 0 \implies \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} X_2 = 0$$

$$X_2 = \begin{bmatrix} -1 & 4 & | & 0 \\ 1 & -4 & | & 0 \end{bmatrix} = \begin{bmatrix} -1 & 4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} = \begin{bmatrix} 1 & -4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \implies X_2 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Hence, the characteristic equation method is verified.

8. Some Important Properties:

- Sum of eigenvalues of a square matrix is equal to its trace and product of the eigenvalues is equal to its determinant.
- For triangular matrices, eigenvalues are its diagonal entries.
- For real symmetric matrices with distinct eigenvalues, the eigenvectors are orthogonal.
- If λ is an eigenvalue of A corresponding to eigenvector X , then λ^n is the eigenvalue of A^n corresponding to X .
- If $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of A then $k\lambda_1, k\lambda_2, \dots, k\lambda_n$ or eigenvalues of kA where $k \neq 0$.
- If λ is an eigenvalue of A , then $\lambda \pm k$ are eigenvalues of $A \pm kI$.
- If λ is an eigenvalue of a non singular matrix A corresponding to eigenvector X , then λ^{-1} is an eigenvalue of A^{-1} corresponding to X .
- If λ is an eigenvalue of a non singular matrix A , then eigenvalue of $\text{adj}(A)$ is $|A|\lambda^{-1}$.
- If λ is an eigenvalue of an orthogonal matrix A , then $\frac{1}{\lambda}$ is also an eigenvalue of the matrix A .
- Eigenvalues of A and A^T are same.

9. If $A = \begin{pmatrix} 8 & -4 \\ 2 & 2 \end{pmatrix}$, determine the eigenvalues and eigenvectors of $B = 2A^2 - \frac{1}{2}A + 3I$.

$$\begin{aligned} |A - \lambda I| = 0 &\implies \begin{vmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{vmatrix} = 0 \implies (8 - \lambda)(2 - \lambda) + 8 = 0 \\ &\implies \lambda^2 - 10\lambda + 24 = 0 \implies \lambda = 6, 4 \end{aligned}$$

Clearly, we know

$$\begin{aligned} \lambda_B &= 2\lambda^2 - \frac{1}{2}\lambda + 3 \\ \therefore \lambda_{B1} &= 2(6)^2 - 3 + 3 \implies \boxed{\lambda_{B1} = 72} \\ \therefore \lambda_{B2} &= 2(4)^2 - 2 + 3 \implies \boxed{\lambda_{B2} = 33} \end{aligned}$$

Also, eigenvectors of B = eigenvectors of A.

Hence,

$$(A - \lambda_1 I)X_1 = 0 \implies \begin{pmatrix} 2 & -4 \\ 2 & -4 \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \implies \boxed{X_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}}$$

$$(A - \lambda_2 I)X_2 = 0 \implies \begin{pmatrix} 4 & -4 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 4 & -4 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \implies \boxed{X_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

1.13 Diagonalization of a Matrix:

1. If a square matrix A of order n does 'n' linearly independent eigenvectors, say $X_1, X_2, \dots, X_n$ corresponding to 'n' eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ respectively, then a matrix S can be found s.t. $S^{-1}AS$ is a diagonal matrix.
2. Let $S = [X_1 \ X_2 \ \dots \ X_n]$
 $\implies AS = [AX_1 \ AX_2 \ \dots \ AX_n]$
 But we know, $AX_1 = \lambda_1 X_1, AX_2 = \lambda_2 X_2, \dots, AX_n = \lambda_n X_n$
 $\therefore AS = [\lambda_1 X_1 \ \lambda_2 X_2 \ \dots \ \lambda_n X_n]$

$$\therefore AS = [X_1 \ X_2 \ \dots \ X_n] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} = S\Lambda$$

Hence,

$$AS = S\Lambda \implies \boxed{S^{-1}AS = \Lambda}$$

where Λ is a diagonal matrix.

3. For real symmetric matrices with distinct eigenvalues, then eigenvectors are orthogonal to each other $\implies S^{-1} = S^T$.

Hence, for real symmetric matrices, $\boxed{S^T AS = \Lambda}$

1.13.1 Power of a Matrix:

We know that $S^{-1}AS = \Lambda \implies A = S\Lambda S^{-1}$

Now, $A^2 = (S\Lambda S^{-1})(S\Lambda S^{-1}) = S\Lambda^2 S^{-1}$

Similarly, $A^3 = (S\Lambda^2 S^{-1})(S\Lambda S^{-1}) = S\Lambda^3 S^{-1}$

Hence, $A^n = S\Lambda^n S^{-1}$

2 Advanced Calculus:

2.1 Precise Definition of a Limit

$$\lim_{x \rightarrow x_0} f(x) = L$$

For a given $\epsilon > 0 \exists \delta > 0$ s.t. $|f(x) - L| < \epsilon$ when $|x - x_0| < \delta$

Now, $|f(x) - L| < \epsilon \implies -\epsilon < f(x) - L < \epsilon \implies f(x) \in (L - \epsilon, L + \epsilon)$
 Also, $|x - x_0| < \delta \implies -\delta < x - x_0 < \delta \implies x \in (x_0 - \delta, x_0 + \delta)$

1. $\epsilon - \delta$ **definition for LHL:**

For a given $\epsilon > 0 \exists \delta > 0$ s.t. $|f(x) - L| < \epsilon$ when $x_0 - \delta < x < x_0$

2. $\epsilon - \delta$ **definition for RHL:**

For a given $\epsilon > 0 \exists \delta > 0$ s.t. $|f(x) - L| < \epsilon$ when $x_0 < x < x_0 + \delta$

Question: Show that $\lim_{x \rightarrow 1} (5x - 3) = 2$.

For a given $\epsilon > 0 \exists \delta > 0$ s.t. $|5x - 3 - 2| < \epsilon$ when $|x - 1| < \delta$

$$\therefore |5x - 5| < \epsilon \implies 5|x - 1| < \epsilon \implies 5\delta = \epsilon \implies \boxed{\delta = \frac{1}{5}\epsilon}$$

2.2 Some Important Definitions:

1. **Sandwich Theorem:** Suppose that $f(x) \leq h(x) \leq g(x)$ for all x in some open interval containing c except possibly at $x = c$ itself.

Then if $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = L \implies \lim_{x \rightarrow c} h(x) = L$

2. **Continuity:** For a function to be continuous, $\lim_{x \rightarrow c} f(x)$ must exist, $f(c)$ must be defined and $\lim_{x \rightarrow c} f(x) = f(c)$

3. **Horizontal Asymptote:** A line $y = b$ is a horizontal asymptote of the graph of the function $y = f(x)$ if $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$

4. **Vertical Asymptote:** A line $x = a$ is a vertical asymptote of the graph of the function $y = f(x)$ if $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ and $\lim_{x \rightarrow a^+} f(x) = \pm\infty$

5. **First Principle of Differentiation:**

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x) - f(x-h)}{h} \right]$$

6. Every differentiable function is continuous.

Proof: Let $f(x)$ be a differentiable function

$$\implies f'(x) = \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} \right]$$

For $f(x)$ to be continuous, $\lim_{x \rightarrow c} f(x) = f(c)$. Consider,

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c}(x - c)$$

Taking limits,

$$\lim_{x \rightarrow c} f(x) - f(c) = f'(c) \lim_{x \rightarrow c} (x - c) = 0$$

Hence, $\lim_{x \rightarrow c} f(x) - f(c) \implies \lim_{x \rightarrow c} f(x) = f(c)$

Hence, proved.

2.2.1 Rolle's Theorem:

Let $f(x)$ be a function that satisfies the following conditions:

1. $f(x)$ is continuous on the closed interval $[a, b]$.
2. $f(x)$ is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$.

Then, there exists a point c in (a, b) such that $f'(c) = 0$

Question: Verify Rolle's Theorem for $f(x) = (x - a)^m(x - b)^n$ on $[a, b]$.

Answer: As $f(x)$ is a polynomial function, it is continuous on $[a, b]$ and is differentiable on (a, b) .

Also, $f(a) = (a - a)^m(a - b)^n = 0$ and $f(b) = (b - a)^m(b - b)^n = 0$

Hence, all conditions of Rolle's Theorem are satisfied.

Now,

$$\begin{aligned} f'(x) &= m(x - a)^{m-1}(x - b)^n + n(x - a)^m(x - b)^{n-1} \\ \implies f'(c) &= (c - a)^{m-1}(c - b)^{n-1}[m(c - b) + n(c - a)] = 0 \\ \implies c &= a, c = b, \boxed{c = \frac{1}{m+n}(na + mb)} \end{aligned}$$

2.2.2 Lagrange's Mean Value Theorem (LMVT):

Let $f(x)$ be a function that satisfies the following conditions:

1. $f(x)$ is continuous on the closed interval $[a, b]$.
2. $f(x)$ is differentiable on the open interval (a, b) .

Then, there exists a point c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

LMVT becomes Rolle's Theorem when $f(b) = f(a)$

2.2.3 Cauchy's Mean Value Theorem (CMVT):

Let $f(x)$ and $g(x)$ be two functions that satisfy the following conditions:

1. $f(x)$ and $g(x)$ are continuous on the closed interval $[a, b]$
2. $f(x)$ and $g(x)$ are differentiable on the open interval (a, b)
3. $g'(c) \neq 0$.

Then, there exists a point c in (a, b) such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

CMVT becomes LMVT when $g(x) = x$.

2.2.4 Increasing and Decreasing Functions:

Suppose that a function $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) such that:

1. If $f'(x) > 0$ at each point $x \in (a, b)$ then f is increasing in $[a, b]$.
2. If $f'(x) < 0$ at each point $x \in (a, b)$ then f is decreasing in $[a, b]$.

2.2.5 First Derivative Test for Local Extrema:

Suppose c is a critical point of a continuous Function $f(x)$ and $f(x)$ is their differentiable at every point of some interval containing c except possibly at c itself then:

1. If $f'(x)$ changes from positive to negative at c , then $f(x)$ has local maxima.
2. If $f'(x)$ changes from negative to positive at c , then $f(x)$ has local minima.
3. If $f'(x)$ changes doesn't change sign at c , then $f(x)$ has inflection point.

The graph of $f(x)$ of a differentiable function $f(x)$ is:

1. concave upward on an open interval I if $f'(x)$ is increasing in I .
2. concave downward on an open interval I if $f'(x)$ is decreasing in I .

2.2.6 Second Derivative Test for Concavity:

Let $f(x)$ be a twice differentiable function on an open interval I , then if

1. $f''(x) > 0$, the graph of $f(x)$ over I is concave up, $f(x)$ attains minima.
2. $f''(x) < 0$, the graph of $f(x)$ over I is concave down, $f(x)$ attains maxima.

- By graph, we can say that if $\Delta = f(x+h) - f(x)$ and
1. $\Delta > 0 \implies f$ attains minima.
 2. $\Delta < 0 \implies f$ attains maxima.

We know that, by Taylor series expansion,

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x)$$

For critical value of x , $f'(x) = 0$.

Hence,

$$\Delta = f(x+h) - f(x) = \frac{h^2}{2!}f''(x)$$

Hence, $\text{sgn}(\Delta) = \text{sgn}(f''(x))$

1. $f''(x) > 0 \implies f$ attains minima.
2. $f''(x) < 0 \implies f$ attains maxima.

2.3 Integration:

1. Integration of a function is the area under the given function in a given interval. We can achieve this by discretely summing the area of infinitesimally small rectangles in the width of the curve.

Hence, Area =

$$\lim_{n \rightarrow \infty} \left[\sum_{i=1}^n f(x_i) \Delta x_i \right] = \int_a^b f(x) dx$$

2. If $y = f(x)$ is non-negative and integrable for a closed interval $[a, b]$ then the area under the curve $y = f(x)$ over $[a, b]$, $A = \int_a^b f(x) dx$
3. If $f(x)$ is continuous on $[a, b]$, then for some point $c \in [a, b]$, the average value of function is $f(c) = \frac{1}{b-a} \int_a^b f(x) dx$
4. **Fundamental Theorem of Calculus - Part 1:** If $f(x)$ is continuous on $[a, b]$, then $F(x) = \int_a^x f(t) dx$ is continuous on $[a, b]$ and differentiable on (a, b) , and its derivative is:

$$\frac{d}{dx} \int_a^x f(t) dx = f(x)$$

5. **Fundamental Theorem of Calculus - Part 2:** If $f(x)$ is continuous on $[a, b]$, then $F(x)$ is any anti-derivative of $f(x)$ on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

6. Area Bounded Between Two Curves:

- (a) If f and g are continuous with $f(x) \geq g(x)$ throughout the closed interval $[a, b]$, then area of region bounded between the curves $y = f(x)$ and $y = g(x)$ from a to b is $A = \int_a^b [f(x) - g(x)]dx$
- (b) If f and g are continuous with $f(y) \geq g(y)$ throughout the closed interval $[c, d]$, then area of region bounded between the curves $x = f(y)$ and $x = g(y)$ from c to d is $A = \int_c^d [f(y) - g(y)]dy$

2.4 Volume of Solid Shape:

The volume of a solid of integrable cross-sectional area A is

$$\lim_{n \rightarrow \infty} \left[\sum_{i=1}^n A(x_i) \Delta x_i \right] = \int_a^b A(x) dx$$

2.4.1 Solid of Revolution - Disk Method:

This method is used to create solids enclosed only by one curve.

$$V_x = \int \pi R^2(x) dx$$

1. **Question:** Region between the curve $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is rotated about x -axis to generate a solid. Find its volume.

Answer:

$$V = \int_0^4 \pi y^2 dx = \int_0^4 \pi x dx = \frac{\pi}{2} x^2 \Big|_0^4 = 8\pi$$

2. **Question:** A circle $x^2 + y^2 = a^2$ is rotated about x -axis to form a sphere. Find its volume.

Answer:

$$V = \int_{-a}^a \pi y^2 dx = 2 \int_0^a \pi (a^2 - x^2) dx = 2\pi \left[a^2 x - \frac{1}{3} x^3 \right] \Big|_0^a = \frac{4}{3} \pi a^3$$

2.4.2 Solid of Revolution - Washer Method:

This method is used to create solids enclosed between two curves.

$$V_x = \int \pi [R^2(x) - r^2(x)] dx$$

Question: Find the volume of the solid enclosed by $y = x^2$ and $y = 2x$ in the First Quadrant about y -axis.

Answer:

$$V = \int_0^4 \pi ((\sqrt{y})^2 - (y/2)^2) dy = \int_0^4 \pi (y - y^2/4) dy$$
$$\therefore V = \pi \left[\frac{y^2}{2} - \frac{y^3}{12} \right] \Big|_0^4 = \frac{8\pi}{3}$$

2.4.3 Solid of Revolution - Shell Method:

The volume of a solid generated by revolving the region between the x-axis and the graph of the continuous function $y = f(x)$ about the vertical line $x = l$ is:

$$V = \int_a^b 2\pi(\text{shell radius})(\text{shell height})dx = \int_a^b 2\pi(x-l)f(x)dx$$

Question: The region bounded by the curve $y = \sqrt{x}$, the x-axis and $x = 4$ is revolved about the line $x = 6$ to generate a solid. Find its volume.

Answer:

$$V = \int_0^4 2\pi(6-x)(\sqrt{x})dx = 2\pi \int_0^4 (6x^{1/2} - x^{3/2})dx$$

$$V = 2\pi \left[4x^{3/2} - \frac{2}{5}x^{5/2} \right]_0^4 = 2\pi \left[32 - \frac{64}{5} \right] = \frac{192\pi}{5}$$

2.5 Arc Length:

1. If $f(x)$ is a continuous function on $[a, b]$, then arc length of $y = f(x)$ from the point $A(a, f(a))$ to point $B(b, f(b))$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

2. If $g(y)$ is a continuous function on $[c, d]$, then arc length of $x = g(y)$ from the point $C(g(c), c)$ to point $D(g(d), d)$ is

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

3. **Question:** Find the arc length of the curve $y = \frac{4}{3}\sqrt{2}x^{3/2} - 1$ for $0 \leq x \leq 1$

Answer:

$$y = \frac{4}{3}\sqrt{2}x^{3/2} - 1 \implies y' = \frac{4}{3}\sqrt{2}x^{1/2} = \sqrt{8x} \implies (y')^2 = 8x$$

$$\therefore L = \int_0^1 \sqrt{1+8x}dx = \left[\frac{1}{8} \frac{2}{3} (8x+1)^{3/2} \right]_0^1 = \frac{13}{6} \text{ units}$$

2.6 Area of Surface of Revolution:

1. If $f(x)$ is a continuously differentiable function on $[a, b]$, then the area of the surface generated by revolving the graph $y = f(x)$ of about the x-axis is

$$A = \int_a^b 2\pi f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

2. If $g(y)$ is a continuously differentiable function on $[c, d]$, then the area of the surface generated by revolving the graph $x = g(y)$ of about the y -axis is

$$A = \int_c^d 2\pi g(y) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

3. **Question:** Find the area of the surface generated by revolving $y = 2\sqrt{x}$ ($1 \leq x \leq 2$) about the x -axis.

Answer:

$$y = 2\sqrt{x} \implies y' = \frac{1}{\sqrt{x}} \implies (y')^2 = \frac{1}{x}$$

$$\therefore A = \int_1^2 2\pi(2\sqrt{x}) \sqrt{1 + \frac{1}{x}} dx = \int_1^2 4\pi\sqrt{x+1} dx = \left[\frac{2}{3}(4\pi)(x+1)^{3/2} \right]_1^2$$

$$\implies A = \frac{8\pi}{3}(3^{3/2} - 2^{3/2}) = 19.835 \text{ units}^2$$

2.7 Center Of Mass:

1. Moment about x -axis (M_x) = $\int \frac{y}{2} dm$
2. Moment about y -axis (M_y) = $\int x dm$
3. Total Mass, $M = \int dm = \int \delta dA$
4. Hence,

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M} \right) = \left(\frac{\int x dm}{\int dm}, \frac{\int (y/2) dm}{\int dm} \right)$$

5. Center of Mass of region lying between the two curves $y = f(x)$ and $y = g(x)$ where $f(x) \geq g(x)$, then:

$$\bar{y} = \frac{1}{M} \int \frac{\delta}{2} (f^2(x) - g^2(x)) dx$$

$$\bar{x} = \frac{1}{M} \int \delta x (f(x) - g(x)) dx$$

where $M = \int \delta (f(x) - g(x)) dx$

6. The triangle enclosed by $y = 2x$, $x = 1$, and the x -axis has a constant density of $\delta = 3gm/cm^2$, then find the x -coordinate of the COM.

$$M = \int dm = \int \delta dA = \int_0^1 3(2x) dx = 3gm$$

$$M_y = \int x dm = \int_0^1 x(6x) dx = 2gmcm$$

$$\therefore \bar{x} = \frac{M_y}{M} = \frac{2gmcm}{3gm} = 2/3cm$$

2.8 Transcendental Functions:

2.8.1 Natural Logarithm ($\ln(x)$):

1. It is a function given by $\ln(x) = \int_1^x \frac{1}{t} dt$ ($x > 0$)
2. By the Fundamental Theorem of Calculus - Part 1:

$$\frac{d}{dx}(\ln(x)) = \frac{d}{dx} \int_1^x \frac{1}{t} dt \implies \boxed{\frac{d}{dx}(\ln(x)) = \frac{1}{x}}$$

3. For any numbers $b, x > 0$, natural logarithm satisfies the following:
 - (a) Product Rule: $\ln(bx) = \ln(b) + \ln(x)$
 - (b) Quotient Rule: $\ln(b/x) = \ln(b) - \ln(x)$
 - (c) Reciprocal Rule: $\ln(1/x) = -\ln(x)$
 - (d) Power Rule: $\ln(x^r) = r \ln(x)$
4. For any numbers $a > 0$ and x , the logarithm function with base a is:

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$

2.8.2 Exponential Function (e^x):

1. For every real number x , we define the natural exponent function as e^x .
2. $e^{\ln(x)} = x$ and $\ln(e^x) = x$
- 3.

$$\int e^x dx = \frac{d}{dx}(e^x) = e^x$$

4. For any numbers $x, x_1, x_2 > 0$, natural exponent satisfies the following:
 - (a) Product Rule: $e^{x_1} e^{x_2} = e^{x_1+x_2}$
 - (b) Quotient Rule: $e^{x_1}/e^{x_2} = e^{x_1-x_2}$
 - (c) Reciprocal Rule: $e^{-x} = \frac{1}{e^x}$
 - (d) Power Rule: $(e^x)^r = e^{rx}$
5. For any numbers $a > 0$ and x , the exponential function with base a is:

$$a^x = e^{x \ln(a)}$$

2.9 Hyperbolic Functions:

1.

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \cosh(x) = \frac{e^x + e^{-x}}{2}$$

2.

$$\frac{d}{dx} \sinh(x) = \cosh(x), \frac{d}{dx} \cosh(x) = \sinh(x)$$

3.

$$\cosh^2(x) - \sinh^2(x) = 1$$

4.

$$\cosh^2(x) + \sinh^2(x) = \cosh(2x)$$

2.10 Gamma Function (Γ):

1. If n is positive, then the gamma function is defined as:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$$

2. $\Gamma(1) = 1$

3.

$$\Gamma(n+1) = \begin{cases} n \cdot \Gamma(n) & , n \in \mathbb{R} - \mathbb{Z} \\ n! & , n \in \mathbb{Z}^+ \end{cases}$$

4.

$$\Gamma(1/2) = \int_0^\infty e^{-x} x^{-1/2} dx$$

Let $x = t^2 \implies dx = 2dt \implies I = \Gamma(1/2) = \int_0^\infty e^{-t^2} 2dt$

Squaring both sides,

$$I^2 = \int_0^\infty 2e^{-x^2} dx \cdot \int_0^\infty 2e^{-y^2} dy$$

$$\therefore I^2 = 4 \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$$

Now,

$$\int_X \int_Y F(x, y) dx dy = \int_R \int_\theta F(r \cos \theta, r \sin \theta) dr d\theta$$

$$= \begin{vmatrix} \delta x / \delta r & \delta x / \delta \theta \\ \delta y / \delta r & \delta y / \delta \theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

$$\therefore I^2 = 4 \int_0^\infty \int_0^{\pi/2} e^{-r^2} r dr d\theta = 4(\pi/2) \int_0^\infty r e^{-r^2} dr$$

Let $r^2 = u \implies 2rdr = du \implies rdr = du/2$

$$I^2 = \pi \int_0^\infty e^{-u} du = -\pi e^{-u} \Big|_0^\infty = \pi$$

Hence,

$$I = \Gamma(1/2) = \sqrt{\pi}$$

5. Non positive integers are not allowed in the Gamma Function.

2.11 Beta Function (β):

1. It is defined as:

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

2.

$$\beta(m, n) = \beta(n, m)$$

3.

$$\beta(m, n) = \begin{cases} 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \\ 2 \int_0^{\pi/2} \sin^{2n-1} \theta \cos^{2m-1} \theta d\theta \end{cases}$$

4.

$$\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

5.

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

6.

$$\beta(m, n) = \beta(m+1, n) + \beta(m, n+1)$$

7. **Question:** Express I in terms of beta function.

Solution: Let $x^p = u \implies px^{p-1} dx = du$
and $x = u^{1/p} \implies x^{p-1} = u^{(p-1)/p} = u^{1-1/p}$

$$\therefore pu^{1-1/p} dx = du \implies dx = \frac{1}{p} u^{\frac{1}{p}-1} du$$

$$I = \int_0^1 (u^{1/p})^m (1-u)^n \left(\frac{1}{p} u^{\frac{1}{p}-1} du \right) \implies I = \frac{1}{p} \int_0^1 u^{\frac{m+1}{p}-1} (1-u)^n du$$

Hence,

$$I = \int_0^1 x^m (1-x^p)^n dx = \frac{1}{p} \beta\left(\frac{m+1}{p}, n+1\right)$$

8.

$$\int_0^{\pi/2} \sin^{2m-1} \theta d\theta = \frac{2m-2}{2m-1} \times \frac{2m-4}{2m-3} \times \dots \times \frac{2}{3} \times 1$$

9.

$$\int_0^{\pi/2} \sin^{2m} \theta d\theta = \frac{2m-1}{2m-0} \times \frac{2m-3}{2m-2} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}$$

10.

$$\beta(x, 1-x) = \frac{\pi}{\sin(\pi x)}$$

11.

$$\Gamma\left(\frac{1}{2} + x\right)\left(\frac{1}{2} - x\right) = \frac{\pi}{\cos(\pi x)}$$

2.12 Limits with two variables:

1. Given limit:

$$\lim_{(x,y) \rightarrow (a,b)} [f(x,y)] = L$$

$\forall \epsilon > 0 \exists \delta > 0$ such that $|f(x,y) - L| < \epsilon$ for $\sqrt{(x-a)^2 + (y-b)^2} < \delta$

2. We can say a function $f(x, y)$ approaches the limit L as $(x, y) \rightarrow (a, b)$.
Mathematically,

$$\lim_{(x,y) \rightarrow (a,b)} [f(x,y)] = L \iff \lim_{x \rightarrow a} [f(x,b)] = L = \lim_{y \rightarrow b} [f(a,y)]$$

3. If $f(x,y) \rightarrow L_1$ as $(x,y) \rightarrow (a,b)$ along the path c_1 and $f(x,y) \rightarrow L_2$ as $(x,y) \rightarrow (a,b)$ along the path c_2 where $L_1 \neq L_2$, then limit L doesn't exist.

4. **Question:** Evaluate the limit

$$L = \lim_{(x,y) \rightarrow (0,0)} \left[\frac{xy^2}{x^2 + y^4} \right]$$

Solution: Along x-axis, $y = 0 \implies L = \lim_{x \rightarrow 0} \left[\frac{0}{x^2} \right] = 0$

Along y-axis, $x = 0 \implies L = \lim_{y \rightarrow 0} \left[\frac{0}{y^4} \right] = 0$

Along $y^2 = mx$,

$$L = \lim_{x \rightarrow 0} \left[\frac{x(mx)}{x^2 + (mx)^2} \right] = \lim_{x \rightarrow 0} \left[\frac{mx^2}{(m^2 + 1)x^2} \right] = \frac{m}{m^2 + 1}$$

As L is dependent of m , the limit L doesn't exist.

2.13 Continuity with two variables:

A function $f(x, y)$ is continuous at a point (x_0, y_0) if:

1. $f(x, y)$ is defined at (x_0, y_0) .

- 2.

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)] \text{ exists.}$$

- 3.

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)] = f(x_0, y_0)$$

2.14 Partial Derivatives:

1. For a function $z = f(x, y)$, the partial derivative $\frac{\partial z}{\partial x}$ (also written as z_x) means that "y" is treated as a constant while differentiating wrt x.

2. **Question:** If $z = x^2 + 3xy + y - 1$, then find z_x and z_y at $(4, -5)$.

Solution:

$$z_x = 2x + 3y \implies z_x|_{(4,-5)} = 2(4) + 3(-5) = -7$$

$$z_y = 3x + 1 \implies z_y|_{(4,-5)} = 3(4) + 1 = 13$$

- 3.

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} \implies z_{xy} = z_{yx}$$

2.14.1 Chain Rule:

1. If $w = f(x)$ is differentiable function of x and $x = g(t)$ is a differentiable function of t , then

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt}$$

2. If $w = f(x, y)$ is differentiable function of x and y , and $x = g(t)$, $y = h(t)$ are differentiable functions of t , then

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

3. If $w = f(x, y, z) = 0$ defines the variable z as a function of $z = z(x, y)$, then:

$$z_x = -\frac{w_x}{w_z}, \quad z_y = -\frac{w_y}{w_z}$$

4. If $u = f(y-z, z-x, x-y)$, then $u_x + u_y + u_z = 0$

5. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, then $xu_x + yu_y + zu_z = 0$

2.15 Extremum of two variables:

Let $z = f(x, y)$ be a function in two variables. Now, if $f(a+h, b+k) < f(a, b)$ then maxima exist and if $f(a+h, b+k) > f(a, b)$ then minima exist.

If $\Delta = f(a+h, b+k) - f(a, b)$, $\Delta > 0 \implies$ minima and $\Delta < 0 \implies$ maxima.

We know,

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2!}h^2f''(x) + \dots$$

$$\therefore f(x+h, y+k) = f(x, y) + (hf_x + kf_y) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f + \dots$$

For extrema, $f_x, f_y = 0$ and let $f_{xx} = r, f_{xy} = s, f_{yy} = t$.

$$\Delta = \frac{1}{2}[h^2r + 2hks + k^2t] = \frac{1}{2r}[h^2r^2 + 2hksr + k^2tr]$$

$$\Delta = \frac{1}{2} \left[\frac{(hr + sk)^2 - s^2k^2 + k^2rt}{r} \right] = \frac{1}{2} \left[\frac{(hr + sk)^2 - k^2(s^2 + rt)}{r} \right]$$

Hence,

1. if $rt - s^2 > 0$ and $r > 0 \implies$ minima
2. if $rt - s^2 > 0$ and $r < 0 \implies$ maxima
3. if $rt - s^2 < 0 \implies$ saddle point

2.16 Lagrange's method of undetermined multipliers:

Suppose that it is required to find the extremum of a function $f(x, y, z) = 0$ subjected to constraints $\phi_1(x, y, z), \phi_2(x, y, z), \dots$, then:

1. Form the Lagrange' function $F(x, y, z)$ as:

$$F(x, y, z) = f(x, y, z) + \lambda_1\phi_1(x, y, z) + \lambda_2\phi_2(x, y, z) + \dots$$

where $\lambda_1, \lambda_2, \dots$ are called "Lagrange' Multipliers".

2. Differentiate wrt each variable and equate to zero. Finally solve for all λ_i and get a relation between x, y, z and solve.

Question: Find the shortest distance from origin to the surface $xyz^2 = 2$

Solution: Let $f = x^2 + y^2 + z^2$ and $\phi = xyz^2 - 2$

$$F(x, y, z) = x^2 + y^2 + z^2 + \lambda(xyz^2 - 2)$$

$$F_x = 2x + \lambda yz = 0 \implies \lambda = \frac{-2x}{yz}$$

$$F_y = 2y + \lambda xz = 0 \implies \lambda = \frac{-2y}{xz}$$

$$F_z = 2z + 2\lambda xyz = 0 \implies \lambda = \frac{-1}{xy}$$

$$\lambda = \frac{-2x}{yz} = \frac{-2y}{xz} = \frac{-1}{xy}$$

Hence, $x^2 = y^2$ and $2y^2 = z^2 \implies 2x^2 = 2y^2 = z^2$

$$xyz^2 = 1 \implies x^4 = 1 \implies x = 1 \implies (1, 1, \sqrt{2})$$

Hence, distance, $d = \sqrt{1^2 + 1^2 + \sqrt{2}^2} = 2$

3 Sequences and Series:

3.1 Sequences:

1. A sequence of real numbers is a set of real numbers arranged in a well defined order. It is denoted by $\{a_n\}$ where a_n is its n^{th} term.
2. **Operations on sequences:** If $\{a_n\}$ and $\{b_n\}$ are two sequences, then:
 - (a) their sum is denoted by $\{a_n + b_n\} = \{a_n\} + \{b_n\}$
 - (b) their product is denoted by $\{a_n \cdot b_n\} = \{a_n\} \cdot \{b_n\}$
 - (c) scalar multiplication is denoted by $\{c \cdot a_n\} = c \cdot \{a_n\}$
3. **Constant Sequences:** A sequence $\{a_n\}$ is denoted by $a_n = k$ is called as a constant sequence ($\{k\}$)
4. A sequence $\{a_n\}$ converges to L if for every positive number ϵ , there exists a corresponding number N such that $\forall \epsilon > 0 \exists n > N$ s.t. $|a_n - L| < \epsilon$
5. If the limit of sequence $\{a_n\}$ exists, then it is unique.
6. If the subsequences $\{a_{2n-1}\}$ and $\{a_{2n+1}\}$ of the sequence $\{a_n\}$ converges to the same limit L, then the sequence $\{a_n\}$ also converges to L.

3.2 Infinite Series:

Given a sequence $\{a_n\}$, an expression of the form:

$$a_1 + a_2 + a_3 + \cdots + a_k + \cdots + a_n + \cdots$$

is an infinite series and the term a_n is the n^{th} term of the series.

3.2.1 Partial Sum Test:

$$s_1 = a_1, s_2 = a_1 + a_2, s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k$$

The sequence $\{s_k\}$ is the sequence of partial sums of the series. The number s_n is the n^{th} partial term. If the sequence of the partial sums converges to a limit L , then we say that the series converges and the sum is L .

$$\lim_{n \rightarrow \infty} s_n = L \implies \sum_{n=1}^{\infty} a_n = L$$

If the n^{th} partial term diverges, then the series also diverges.

3.2.2 Limit Test:

1. If $\sum_{n=1}^{\infty} a_n$ converges then limit $\lim_{n \rightarrow \infty} a_n = 0$.
2. If $\sum_{n=1}^{\infty} a_n$ diverges then limit $\lim_{n \rightarrow \infty} a_n$ fails to exist or is not zero.

3.2.3 Adding / Subtracting Terms:

1. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum a_n + \sum b_n$ also diverges.
2. If $\sum a_n$ and $\sum b_n$ both diverges, then we cannot predict the nature of $\sum a_n \pm \sum b_n$.
3. If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=k}^{\infty} a_n$ is convergent for $k > 1$.

3.2.4 Integral Test:

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$ where f is a continuous, positive, decreasing function of n , then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(n)dn$ both converges or diverges together.

3.2.5 p-Series:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \dots$$

The above series is called the "p-series", and it converges only if $p > 1$.

3.2.6 Geometric Series:

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \dots$$

The above series is called the "Geometric Series", and it converges only if $|r| < 1$.

3.2.7 Comparison Test:

Let $\sum a_n$, $\sum d_n$, $\sum c_n$ are three series with non-negative terms. Suppose that for some N , $d_n \leq a_n \leq c_n \forall N$, then:

1. If $\sum c_n$ converges, then $\sum a_n$ also converges.
2. If $\sum d_n$ diverges, then $\sum a_n$ also diverges.

3.2.8 Limit Comparison Test:

Suppose that $a_n > 0$ and $b > 0 \forall n \geq N$, then:

1. if $\lim_{n \rightarrow \infty} \left[\frac{a_n}{b_n} \right] = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converges/diverges.
2. if $\lim_{n \rightarrow \infty} \left[\frac{a_n}{b_n} \right] = 0$ and $\sum b_n$ converges, then $\sum a_n$ also converges.
3. if $\lim_{n \rightarrow \infty} \left[\frac{a_n}{b_n} \right] = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ also diverges.